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NARAYANA EDUCATIONAL SOCIETY

STAR SUPER CHAINA CAMPUS

Plot no:38,39 & 43,44 MILANO TOWER,KHANAMET HITEX ROAD,MADHAPUR,HYD-84

Sec: IJR (JI-0& JI-1) **ACHIEVER'S QUIZ** **Date:13.07.24**

Topic: DETERMINANTS **Sub :MATHS**

TIME: 1 HR **MARKS:**

SINGLE ANSWER TYPE:

1. Let α,β,γ be the roots of the cubic $x^3+ax^2+bx+c=0$, which (taken in given order) are in G.P. If α and β are such that $\begin{vmatrix} 2 & 1 & 2 \\ 1+\alpha & \alpha & \beta \\ 4-\beta & 3-\beta & \alpha+1 \end{vmatrix}=0$. If $S=\sum_{r=1}^{100}\left(\left(\frac{\alpha}{\beta}\right)^r+\left(\frac{a}{b}\right)^r\right)$, then S equals:

A) $\frac{1}{3}\left(1-\frac{1}{2^{100}}\right)$ B) $\frac{4}{3}\left(1-\frac{1}{2^{100}}\right)$ C) $\frac{8}{3}\left(1-\frac{1}{2^{100}}\right)$ D) $\frac{2}{3}\left(1-\frac{1}{2^{100}}\right)$

2. If $A=\begin{bmatrix} a & b & c \\ x & y & z \\ p & q & r \end{bmatrix}$ and $B=\begin{bmatrix} q & -b & y \\ -p & a & -x \\ r & -c & z \end{bmatrix}$, then

A) $|A|=|B|$ B) $|A|=-|B|$ C) $|A|=2|B|$ D) $|A|=3|B|$

3. Let $\Delta=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ If D_1,D_2,D_3 are cofactors of c_1,c_2 and c_3 respectively such that $D_1^2+D_2^2+D_3^2=16$ and $C_1^2+C_2^2+C_3^2=4$, then maximum value of Δ is k is less than

A) 5 B) 7 C) 8 D) 16

MULTI ANSWER TYPE:

4. Let $f(x)=\begin{vmatrix} \sec^2 x & 1 & 1 \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cot^2 x \end{vmatrix}$, then which of the following statement(s) is/are true?

A) period of $f(x)$ is π

B) maximum value of $f(x)$ is 1

C) minimum value of $f(x)$ is 0

D) product of maximum and minimum values of $f(x)$ is 0

5. Let $R_n=|b_{ij}|_{n\times n}$ be a determinant such that $b_{ij}=\begin{cases} 1, & |i-j|=1 \\ 3, & |i-j|=0 \\ 0, & otherwise \end{cases}$ then $n\in N$

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A) $R_{2019}+R_{2017}=3R_{2018}$ B) $R_{2019}+R_{2017}=9R_{2018}$

C) The value of R_4 is 54 D) R_5 is multiple of 9

6. The value of the determinant $\begin{vmatrix} \cos(\theta+\alpha) & -\sin(\theta+\alpha) & \cos 2\alpha \\ \sin\theta & \cos\theta & \sin\alpha \\ -\cos\theta & \sin\theta & \lambda\cos\alpha \end{vmatrix}$ is

A) independent of θ for all $\lambda \in R$

B) independent of θ and α when $\lambda = 1$

C) independent of θ and α when $\lambda = -1$

D) independent of λ for all θ

7. The determinant $\begin{vmatrix} \sin^2\alpha & \sin\alpha.\cos\alpha & \cos^2\alpha \\ \sin^2\beta & \sin\beta.\cos\beta & \cos^2\beta \\ \sin^2\gamma & \sin\gamma.\cos\gamma & \cos^2\gamma \end{vmatrix}$ is/are equal to

A) $-\pi\sin(\alpha-\beta)$ B) $\sum\sin(\alpha-\beta).\cos\alpha.\cos\beta$

C) $\frac{1}{4}\sum\sin(2\alpha-2\beta)$ D) $\pi\sin(\alpha-\beta)$

NUMERICAL TYPE:

8. If $a=\cot 80^0, b=\cot 60^0$ and $c=\cot 40^0$, then the value of $\begin{vmatrix} -bc & b^2+bc & c^2+bc \\ a^2+ac & -ac & c^2+ac \\ a^2+ab & b^2+ab & -ab \end{vmatrix}$ is equal to

9. Let $A=(a_{ij})_{3\times3}$ where $a_{ij}=i-j$ if $i\times j$ is odd

$=i^2$ if $i\times j$ is even

b_{ij} is cofactor of a_{ij} in A. $d_{ij}=\sum_{k=1}^{n}a_{ik}b_{ki}$ then $\left(\frac{\sqrt[3]{\det(d_{ij})}}{-8}\right)=$ ___

10. Let a third order determinant $\Delta_1=\{a_{ij}\}; i,j\in\{1,2,3\}$ and the determinant Δ_2 is constructed by multiplying the element a_{ij} of Δ_1 by 2^{i-j}, i.e., $\Delta_2=\{2^{i-j}a_{ij}\}; i,j\in\{1,2,3\}$, then $\frac{5\Delta_2+4\Delta_1}{5\Delta_2-4\Delta_1}$ is equal to____

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11. If $a,b,c,d>0;\, x\in R$ and $\left(a^2+b^2+c^2\right)x^2-2\left(ab+bc+cd\right)x+b^2+c^2+d^2\le 0$, then $\begin{vmatrix} 33 & 14 & \ln a \\ 65 & 27 & \ln b \\ 97 & 40 & \ln c \end{vmatrix}$ is equal to :

12. If $\Delta_r=\begin{vmatrix} r & r-1 \\ r-1 & r \end{vmatrix}$ where r is a natural value of $\sqrt[10]{\sum_{r=1}^{1024}\Delta_r}$ is

13. Given that $\begin{vmatrix} a & d & 1 \\ b & e & 1 \\ c & f & 1 \end{vmatrix}=-5,\ \begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ d & e & f \end{vmatrix}=-3$ find the value of $\begin{vmatrix} c & 3 & f \\ b & 1 & e \\ a & -1 & d \end{vmatrix}$.

14. Let $f(x)=\ln\left(x+\sqrt{x^2+1}\right)$, then the value of determinant

determinat $\begin{vmatrix} f(\sin 2017\pi) & f\left(\sin\frac{\pi}{6}\right) & f\left(e^{\pi}\right) \\ f\left(\cos\frac{2\pi}{3}\right) & f\left(\cos\frac{2017\pi}{2}\right) & f\left(\tan\frac{\pi}{3}\right) \\ f\left(-e^{\pi}\right) & f\left(\cot\frac{5\pi}{6}\right) & f(0) \end{vmatrix}$ is

15. If $a_1,a_2,a_3,\ldots\ldots,a_{12}$ are in A.P.(all are distinct) and $\Delta_1=\begin{vmatrix} a_1a_5 & a_1 & a_2 \\ a_2a_6 & a_2 & a_3 \\ a_3a_7 & a_3 & a_4 \end{vmatrix},\Delta_2=\begin{vmatrix} a_2a_{10} & a_2 & a_3 \\ a_3a_{11} & a_3 & a_4 \\ a_4a_{12} & a_4 & a_5 \end{vmatrix}$,

then $\Delta_1:\Delta_2$ is equal to________

16. Consider a matrix 'A' of order 3×3 given by

$A=\begin{vmatrix} x^{2018} & x^{2019}+2018 & x^{2020}-2019 \\ x^{2020}-2018 & x^{2019} & x^{2018}+2020 \\ x^{2019}+2019 & x^{2018}-2020 & x^{2020} \end{vmatrix}, x\in R \text{ and } f(x)=\det(A)$, then the value

of $\dfrac{\prod_{r=1}^{2021} f\left(r(r-2019)\right)}{2018}$ is

17. If $f(x)=\begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$, then $f(\theta)-2f(\phi)+f(\psi)$ is equal to ______

18. If a,b,c are the roots of the equation $x^3-3x^2+3x+7=0$, then the value of

$\begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ac-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix}$ is

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19. If $\begin{vmatrix} x^n & x^{n+2} & x^{n+4} \\ y^n & y^{n+2} & y^{n+4} \\ z^n & z^{n+2} & z^{n+4} \end{vmatrix} = \left(\frac{1}{y^2}-\frac{1}{x^2}\right)\left(\frac{1}{z^2}-\frac{1}{y^2}\right)\left(\frac{1}{x^2}-\frac{1}{z^2}\right)$ then the absolute value of n is

20. The value of the determinant $\begin{vmatrix} \sin 2010\pi & \sin\frac{\pi}{5}\sin\frac{2\pi}{5} & \tan\frac{667\pi}{4} \\ \sin\frac{3\pi}{5} & \cos 2009\frac{\pi}{2} & \cos 2013\frac{\pi}{2} \\ \tan\frac{3\pi}{10} & \cot 2011\frac{\pi}{2} & \sin\frac{4\pi}{5} \end{vmatrix}$ is $\frac{-K}{16}$ Where K is

21. Consider a matrix 'A' of order 3×3 given by $A = \begin{bmatrix} x^{2018} & x^{2019}+2018 & x^{2020}-2019 \\ x^{2020}-2018 & x^{2019} & x^{2018}+2020 \\ x^{2019}+2019 & x^{2018}-2020 & x^{2020} \end{bmatrix}, x\in R$ and $f(x)=\det(A)$, then the value of $\frac{\prod_{r=1}^{2021} f\left(r(r-2019)\right)}{2018}$ is

22. $\begin{vmatrix} 3x^2 & x^2+x\cos\theta+\cos^2\theta & x^2+x\sin\theta+\sin^2\theta \\ x^2+x\cos\theta+\cos^2\theta & 3\cos^2\theta & 1+\sin\theta\cos\theta \\ x^2+x\sin\theta+\sin^2\theta & 1+\sin\theta\cos\theta & 3\sin^2\theta \end{vmatrix} = 0$ Given the real roots of the above determinant are $f(\theta)$ and $g(\theta)$, the value of $\left[f\left(\frac{\pi}{11}\right)\right]^2 + \left[g\left(\frac{\pi}{11}\right)\right]^2$ is ______

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KEY

1	2	3	4	5	6	7	8	9	10
D	**B**	**D**	**ABCD**	**AD**	**AC**	**ABC**	**1**	**6**	**9**
11	12	13	14	15	16	17	18	19	20
0	**4**	**21**	**0**	**1**	**0**	**0**	**0**	**4**	**5**
21	22								
0	**1**								